



A-level FURTHER MATHEMATICS 7367/3S

Paper 3 Statistics

Mark scheme

June 2025

Version: 1.0 Final



Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

No student should be disadvantaged on the basis of their gender identity and/or how they refer to the gender identity of others in their exam responses.

A consistent use of 'they/them' as a singular and pronouns beyond 'she/her' or 'he/him' will be credited in exam responses in line with existing mark scheme criteria.

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Mark scheme instructions to examiners

General

The mark scheme for each question shows:

- the marks available for each part of the question
- the total marks available for the question
- marking instructions that indicate when marks should be awarded or withheld including the principle on which each mark is awarded. Information is included to help the examiner make his or her judgement and to delineate what is creditworthy from that not worthy of credit
- a typical solution. This response is one we expect to see frequently. However credit must be given on the basis of the marking instructions.

If a student uses a method which is not explicitly covered by the marking instructions the same principles of marking should be applied. Credit should be given to any valid methods. Examiners should seek advice from their senior examiner if in any doubt.

Key to mark types

M	mark is for method
R	mark is for reasoning
A	mark is dependent on M marks and is for accuracy
B	mark is independent of M marks and is for method and accuracy
E	mark is for explanation
F	follow through from previous incorrect result

Key to mark scheme abbreviations

CAO	correct answer only
CSO	correct solution only
ft	follow through from previous incorrect result
'their'	indicates that credit can be given from previous incorrect result
AWFW	anything which falls within
AWRT	anything which rounds to
ACF	any correct form
AG	answer given
SC	special case
OE	or equivalent
NMS	no method shown
PI	possibly implied
sf	significant figure(s)
dp	decimal place(s)
ISW	Ignore Subsequent Workings

Examiners should consistently apply the following general marking principles:

No Method Shown

Where the question specifically requires a particular method to be used, we must usually see evidence of use of this method for any marks to be awarded.

Where the answer can be reasonably obtained without showing working and it is very unlikely that the correct answer can be obtained by using an incorrect method, we must award **full marks**. However, the obvious penalty to candidates showing no working is that incorrect answers, however close, earn **no marks**.

Where a question asks the candidate to state or write down a result, no method need be shown for full marks.

Where the permitted calculator has functions which reasonably allow the solution of the question directly, the correct answer without working earns **full marks**, unless it is given to less than the degree of accuracy accepted in the mark scheme, when it gains **no marks**.

Otherwise we require evidence of a correct method for any marks to be awarded.

Diagrams

Diagrams that have working on them should be treated like normal responses. If a diagram has been written on but the correct response is within the answer space, the work within the answer space should be marked. Working on diagrams that contradicts work within the answer space is not to be considered as choice but as working, and is not, therefore, penalised.

Work erased or crossed out

Erased or crossed out work that is still legible and has not been replaced should be marked. Erased or crossed out work that has been replaced can be ignored.

Choice

When a choice of answers and/or methods is given and the student has not clearly indicated which answer they want to be marked, mark positively, awarding marks for all of the student's best attempts. Withhold marks for final accuracy and conclusions if there are conflicting complete answers or when an incorrect solution (or part thereof) is referred to in the final answer.

AS/A-level Maths/Further Maths assessment objectives

AO		Description
AO1	AO1.1a	Select routine procedures
	AO1.1b	Correctly carry out routine procedures
	AO1.2	Accurately recall facts, terminology and definitions
AO2	AO2.1	Construct rigorous mathematical arguments (including proofs)
	AO2.2a	Make deductions
	AO2.2b	Make inferences
	AO2.3	Assess the validity of mathematical arguments
	AO2.4	Explain their reasoning
	AO2.5	Use mathematical language and notation correctly
AO3	AO3.1a	Translate problems in mathematical contexts into mathematical processes
	AO3.1b	Translate problems in non-mathematical contexts into mathematical processes
	AO3.2a	Interpret solutions to problems in their original context
	AO3.2b	Where appropriate, evaluate the accuracy and limitations of solutions to problems
	AO3.3	Translate situations in context into mathematical models
	AO3.4	Use mathematical models
	AO3.5a	Evaluate the outcomes of modelling in context
	AO3.5b	Recognise the limitations of models
	AO3.5c	Where appropriate, explain how to refine models

Q	Marking Instructions	AO	Marks	Typical Solution
1	Circles the 4th answer.	1.1b	B1	6400
Question total			1	

Q	Marking Instructions	AO	Marks	Typical Solution
2	Circles the 1st answer.	2.2a	B1	–6
Question total			1	

Q	Marking Instructions	AO	Marks	Typical Solution
3	Ticks the 1st box.	1.2	B1	He could have made a Type I error but not a Type II error.
Question total			1	

Q	Marking Instructions	AO	Marks	Typical Solution
4(a)	Obtains $\sin(3 \times 0.3) - \sin(3 \times 0.1)$ PI Condone missing brackets or poor bracketing eg $\sin 3(0.3) - \sin 3(0.1)$	1.1a	M1	$F(0.3) - F(0.1)$ $= \sin(3 \times 0.3) - \sin(3 \times 0.1)$ $= 0.488$
	Obtains AWRT 0.488	1.1b	A1	
	Subtotal		2	

Q	Marking Instructions	AO	Marks	Typical Solution
4(b)	Obtains $\sin(3 \times \text{median}) = 0.5$	1.1a	M1	$\sin(3m) = 0.5$ $3m = \frac{\pi}{6}$ $m = \frac{\pi}{18}$
	Completes reasoned argument to show median = $\frac{\pi}{18}$ Must see $3 \times \text{median} = \frac{\pi}{6}$ before final answer	2.1	R1	
	Subtotal		2	

Q	Marking Instructions	AO	Marks	Typical Solution
4(c)	Obtains $3 \cos(3x)$	1.1a	M1	$f(x) = \begin{cases} 3\cos(3x) & 0 \leq x \leq \frac{\pi}{6} \\ 0 & \text{otherwise} \end{cases}$
	Obtains correct fully defined function for $f(x)$ including $f(x) =$	1.1b	A1	
	Subtotal		2	

	Question total		6	
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Q	Marking Instructions	AO	Marks	Typical Solution
5(a)	Obtains $\frac{n^2-1}{12}$ or $\frac{(n+1-n)^2}{12}$ OE	1.1a	M1	$\text{Var}(D) = \frac{n^2-1}{12}$
	Obtains $\frac{n^2-1}{12}$ and $\frac{(n+1-n)^2}{12}$ OE	1.1b	A1	$\text{Var}(R) = \frac{(n+1-n)^2}{12} = \frac{1}{12}$ $\text{Var}(D+R) = \frac{n^2-1}{12} + \frac{1}{12}$
	Completes reasoned argument to obtain $\text{Var}(D+R) = \frac{n^2}{12}$ Var(D) and Var(R) do not need to be explicitly identified but the addition needs to be seen before the simplified final answer An unsimplified Var(R) in terms of n needs to be seen	2.1	R1	$\text{Var}(D+R) = \frac{n^2}{12}$
Subtotal			3	

Q	Marking Instructions	AO	Marks	Typical Solution
5(b)	Obtains $E(D+R) = \frac{n+1}{2} + \frac{n+(n+1)}{2}$ OE PI	1.1b	B1	$E(D+R) = \frac{n+1}{2} + \frac{n+(n+1)}{2}$ $E(D+R) = \frac{3n}{2} + 1$
	Forms an equation using their $E(D+R) = \frac{n^2+44}{12}$ Their $E(D+R)$ must be labelled or correct	1.1a	M1	$\frac{3n}{2} + 1 = \frac{n^2+44}{12}$ $\frac{n^2}{12} - \frac{3n}{2} + \frac{8}{3} = 0$ $2n^2 - 36n + 64 = 0$ $n = 2, 16$ $n = 16$
	Completes reasoned argument by solving a correct equation to obtain $n = 16$ $n = 2$ must be seen and rejected Allow any or missing reason for rejection	2.1	R1	
Subtotal			3	

Question total			6	
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Q	Marking Instructions	AO	Marks	Typical Solution
6(a)	Obtains $\bar{x} = \frac{1003.2}{11}$ OE PI	1.1b	B1	$\bar{x} = \frac{1003.2}{11} = 91.2$ $s^2 = \frac{1}{10} \left(91747.84 - \frac{1003.2^2}{11} \right)$ $s^2 = 25.6$ $t_{10}(95\%) = 1.812$ $91.2 \pm 1.812 \times \sqrt{\frac{25.6}{11}}$ $(88.44, 93.96)$
	Obtains $s^2 = 25.6$ OE or $s =$ AWRT 5.06 PI	1.1b	B1	
	Obtains $t_{10}(95\%) =$ AWRT 1.81 PI	2.2a	B1	
	Evaluates their $\bar{x} + \text{their } t \times \frac{\text{their } s}{\sqrt{11}}$ or $\bar{x} - \text{their } t \times \frac{\text{their } s}{\sqrt{11}}$ where t is a value between 0 and 4 Condone z value Condone σ for s PI	1.1a	M1	
	Obtains correct confidence interval. AWRT 88.44 and 93.96 Allow AWRT 88.43 and 93.97 only if their t is stated as 1.8125	1.1b	A1	
	Subtotal		5	

Q	Marking Instructions	AO	Marks	Typical Solution
6(b)	States both hypotheses using correct language	2.5	B1	$H_0: \mu = 90$ $H_1: \mu \neq 90$ 90 is in the confidence interval Do not reject H_0 Insufficient evidence to suggest that the mean satisfaction score for the university is not 90
	Evaluates the t model by correctly identifying whether 90 lies within their confidence interval and makes a correct corresponding conclusion whether to reject H_0 or Correctly calculates t test statistic AWRT 0.8, compares it to their t value and makes a correct corresponding conclusion whether to reject H_0 OE Condone Accept H_0 /Reject H_1	3.5a	M1	
	Concludes, from a fully correct comparison, in context by referring to mean score (The conclusion must not be definite. eg use of suggest/support) Note an award of B0 M1 R1 is possible	2.2b	R1	
	Subtotal		3	
	Question total		8	

Q	Marking Instructions	AO	Marks	Typical Solution															
7(a)	Obtains correct probability distribution table or function for X or X^n PI	3.3	M1	<table><tr><td>x</td><td>0</td><td>10</td><td>100</td></tr><tr><td>x^n</td><td>0</td><td>10^n</td><td>100^n</td></tr><tr><td>$P(X = x)$</td><td>0.59</td><td>0.37</td><td>0.04</td></tr></table>				x	0	10	100	x^n	0	10^n	100^n	$P(X = x)$	0.59	0.37	0.04
	x	0	10	100															
	x^n	0	10^n	100^n															
	$P(X = x)$	0.59	0.37	0.04															
	Uses their discrete random variable model to obtain $E(X)$ or $E(X^n)$	3.4	M1	$E(X) = 0 \times 0.59 + 10 \times 0.37 + 100 \times 0.04$ $E(X) = 7.7$															
	Obtains $E(X) = 10 \times 0.37 + 100 \times 0.04$ and $E(X^n) = 10^n \times 0.37 + 100^n \times 0.04$ OE	1.1b	A1	$E(X^n) = 10^n \times 0.37 + 100^n \times 0.04$ $0.37(10^n) + 0.04(100^n) = 50 \times 7.7$ $0.04(10^n)^2 + 0.37(10^n) - 385 = 0$															
	Forms equation their $E(X^n) = 50 \times$ their $E(X)$	3.1b	M1	$10^n = 93.59$ $n = \log_{10} 93.59$ $n = 1.97$															
	Obtains correct quadratic equation for their 7.7 in terms of 10^n $0.04(10^n)^2 + 0.37(10^n) = 385$ OE PI by sight of AWRT 94 or AWRT 2	1.1a	M1	New scores = 0, 94, 8759															
Obtains $10^n =$ AWRT 94 PI	1.1b	A1																	
Obtains the scores for the new edition of the game, 0, 94, 8759	3.2a	A1																	
	Subtotal		7																

Q	Marking Instructions	AO	Marks	Typical Solution
7(b)	Refines, or describes how to refine, the correct scores from part (a) to give scores with a mean value at least 385 and all scores are multiples of 10 eg round all the scores up to the nearest 10 or round n to 2 Note for 0, 90, 8760 award E0 but for e.g. 10, 90, 8760 award E1	3.5c	E1	New scores = 0, 100, 8760
	Subtotal		1	
	Question total		8	

Q	Marking Instructions	AO	Marks	Typical Solution									
8	States both hypotheses using correct language OE . Variables need to be stated in at least the null hypothesis.	2.5	B1	H_0 : There is no association between favourite coffee shop and favourite restaurant. H_1 : There is an association between favourite coffee shop and favourite restaurant.									
	Obtains correct observed or expected frequencies Allow expected frequencies to be rounded to 3 s.f. PI	3.3	M1	<table border="1"><tr><td><i>O</i></td><td><i>C</i></td><td><i>D</i></td></tr><tr><td><i>A</i></td><td>52</td><td>78</td></tr><tr><td><i>B</i></td><td>88</td><td>82</td></tr></table>	<i>O</i>	<i>C</i>	<i>D</i>	<i>A</i>	52	78	<i>B</i>	88	82
	<i>O</i>	<i>C</i>	<i>D</i>										
	<i>A</i>	52	78										
	<i>B</i>	88	82										
	Obtains correct observed and expected frequencies Allow expected frequencies to be rounded to 3 s.f. PI	1.1b	A1	<table border="1"><tr><td><i>E</i></td><td><i>C</i></td><td><i>D</i></td></tr><tr><td><i>A</i></td><td>$60\frac{2}{3}$</td><td>$69\frac{1}{3}$</td></tr><tr><td><i>B</i></td><td>$79\frac{1}{3}$</td><td>$90\frac{2}{3}$</td></tr></table>	<i>E</i>	<i>C</i>	<i>D</i>	<i>A</i>	$60\frac{2}{3}$	$69\frac{1}{3}$	<i>B</i>	$79\frac{1}{3}$	$90\frac{2}{3}$
	<i>E</i>	<i>C</i>	<i>D</i>										
<i>A</i>	$60\frac{2}{3}$	$69\frac{1}{3}$											
<i>B</i>	$79\frac{1}{3}$	$90\frac{2}{3}$											
Calculates test statistic $\sum \frac{(O - E - 0.5)^2}{E}$ or $\sum \frac{(O - E)^2}{E}$ Condone missing modulus signs OE PI	3.4	M1	$\sum \frac{(O - E - 0.5)^2}{E} = \frac{\left(\left 52 - 60\frac{2}{3}\right - 0.5\right)^2}{60\frac{2}{3}}$										
Obtains correct value of $\sum \frac{(O - E - 0.5)^2}{E} = \textbf{AWRT 3.6}$	1.1b	A1	$+ \frac{\left(\left 78 - 69\frac{1}{3}\right - 0.5\right)^2}{69\frac{1}{3}} +$										
Obtains correct critical value AWRT 3.8 or obtains $p = \textbf{AWRT 0.056}$	2.2a	B1	$\frac{\left(\left 88 - 79\frac{1}{3}\right - 0.5\right)^2}{79\frac{1}{3}} +$ $\frac{\left(\left 82 - 90\frac{2}{3}\right - 0.5\right)^2}{90\frac{2}{3}}$ $= 3.638$ χ^2 cv for 1 df = 3.841										
Evaluates χ^2 -test statistic by comparing their test statistic with AWRT 3.8 or 5.0 or their p (a test statistic must have been calculated) with 0.05 and makes a correct corresponding conclusion whether to reject H_0 Condone Accept H_0 or reject H_1	3.5a	M1	$3.638 < 3.841$ Do not reject H_0 Insufficient evidence to suggest that there is an association between favourite coffee shop and favourite restaurant.										

	Concludes, from a fully correct comparison, in context by referring to association between coffee shop and restaurant OE (The conclusion must not be definite. eg use of suggest/support)	2.2b	R1	
	Question total		8	

Q	Marking Instructions	AO	Marks	Typical Solution
9(a)	States or uses $X \sim \text{Exp}(60)$, $\text{Exp}(120)$ or $\text{Exp}(2)$ PI by use of correct pdf	3.3	M1	$X \sim \text{Exp}(60)$
	Obtains $\int \lambda x e^{-\lambda x} (dx)$, or $\int \frac{1}{\lambda} x e^{-\frac{1}{\lambda} x} (dx)$ for their λ	3.4	M1	$E(X) = \lim_{n \rightarrow \infty} \left(\int_0^n 60 x e^{-60x} dx \right)$ $= \lim_{n \rightarrow \infty} \left(\left[-x e^{-60x} \right]_0^n - \int_0^n -e^{-60x} dx \right)$ $= \lim_{n \rightarrow \infty} \left(\left[-x e^{-60x} - \frac{1}{60} e^{-60x} \right]_0^n \right)$
	Obtains $-x e^{-\lambda x} - \int -e^{-\lambda x} (dx)$ or $-x e^{-\frac{1}{\lambda} x} - \int -e^{-\frac{1}{\lambda} x} (dx)$ for their λ PI The two terms may be seen on different lines	1.1a	M1	$= \left(-n \times 0 - \frac{1}{60} \times 0 \right)$ $- \left(-0 \times e^{-60 \times 0} - \frac{1}{60} \times e^{-60 \times 0} \right)$ $= \frac{1}{60} \text{ of 30 minutes}$
	Obtains $-x e^{-\lambda x} - \frac{1}{\lambda} e^{-\lambda x}$ for their λ The two terms may be seen on different lines	1.1b	A1F	Mean time = $\frac{1}{60} \times 30$
	Applies correct limits to correctly obtain correct value of the mean $\frac{1}{\lambda}$ where $\lambda = 60, 120$ or 2 Condone lack of limiting process.	1.1b	A1	= 0.5 minutes
	Completes reasoned argument to find that the mean time between text messages being sent is 0.5 Do not condone the lack of a limiting process nor missing dx For $\lambda = 60$, need to see $\frac{1}{60} \times 30$ For $\lambda = 120$, need to see $\frac{1}{120} \times 60$ For $\lambda = 2$, need to see $\lambda = \frac{60}{30}$ or 60 per 30 min cancelled to 2 per min	2.1	R1	
Subtotal			6	

Q	Marking Instructions	AO	Marks	Typical Solution
9(b)	Use Poisson model $Y \sim \text{Po}(120)$ to calculate one of $P(Y > 130) = \text{AWRT } 0.1685$ $P(Y \leq 130) = \text{AWRT } 0.83$ $P(Y \geq 130) = \text{AWRT } 0.19$ $P(Y < 130) = \text{AWRT } 0.81$ Condone incorrect labelling	3.4	M1	$Y \sim \text{Po}(120)$ $P(Y > 130) = 1 - P(Y \leq 130)$ $P(Y > 130) = 1 - 0.8315$ $P(Y > 130) = 0.1685$
	Completes reasoned argument to find that the probability that the company sends more than 130 texts in an hour is 0.169, correct to three decimal places. Must see either a more accurate value for $P(Y > 130) = \text{AWRT } 0.1685$ or $P(Y \leq 130) = \text{AWRT } 0.8315$, condone truncated 0.8314 provided incorrect 0.1686 not also seen, before the final answer with correct labelling OE The Poisson model used, $\text{Po}(120)$, must be stated.	2.1	R1	$P(Y > 130) = 0.169$
Subtotal			2	

Q	Marking Instructions	AO	Marks	Typical Solution
9(c)	States or uses $A \sim B(7, 0.169)$ Allow the use of a more accurate value for 0.169	3.3	M1	$A \sim B(7, 0.169)$ $P(A = 2) =$
	Calculates $\binom{7}{2} \times 0.169^2 \times (1 - 0.169)^{7-2}$ PI	3.4	M1	$\binom{7}{2} \times 0.169^2 \times (1 - 0.169)^{7-2}$ $P(A = 2) = 0.24$
	Obtains $P(A = 2) = \text{AWRT } 0.24$ Do not ignore subsequent working	1.1b	A1	
Subtotal			3	

Question total			11	
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Paper total			50	
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